Evolution Strategies Learning With Variable Impedance Control for Grasping Under Uncertainty

Yingbai Hu, Xinyu Wu, *Member, IEEE*, Peng Geng, and Zhijun Li, *Senior Member, IEEE*

***Abstract*—During a robot's interaction with the environment, it is necessary to ensure the safety and robustness of the robot's movements. To improve the safety and adaptiveness of robots in performing complex movement tasks, a novel method called covariance matrix adaptation-evolution strategies (CMA-ES) for learning complex and high-dimensional motor skills is presented. Considering the complex motion model of trajectories, dynamic movement primitives (DMPs), which is a generic method for trajectories modeling in attractor landscape based on differential dynamic systems, is used to represent the robot's trajectories. CMA-ES offers a theoretical rule for updating the parameters of DMPs and a variable impedance controller, which can reduce the impact of noisy environment on the robot's movement. In this paper, we propose two hierarchies for controlling the robot: the high-level neural-dynamic network optimization for redundancy resolution in task space and the low-level CMA-ES fusing with DMPs for learning trajectories in joint space. In this paper, CMA-ES method is explored to learn variable impedance control and the performance of the proposed method in learning the robot's movements is also tested.**

***Index Terms*—Covariance matrix adaptation-evolution strategies, dynamic movement primitives, redundancy resolution, variable impedance control.**

I. Introduction

In RECENT years, several traditional control methods have been applied to the operation of robots achieving good control performance with fixed model in certain environments. However, the model of robot is typically complex, because the dynamics of a robot is constrained by closed chain mechanism, and meanwhile the system of robot is nonlinear and coupled. Moreover, perturbations can also impact the control system, which brings many challenges to robot's operation. An undesired movement of robot may be dangerous, e.g., casualties and property may be caused under high-frequency perturbations. Therefore, it is crucial to ensure the safety, human-friendliness, and adaptability of a robot's operation, especially when the robot is cooperating with humans or interacting with unpredictable environment [1]–[3].

Traditional impedance control is a general control method that describes the relationship between positions and forces when a robot has physical interactions with the environment and delivers friendly and safe interactive performance [4]. The robot behaves as the mass-spring-dashpot model, while the proposed method is adopted to control it. However, compared with human–robot cooperation, humans cannot keep up with the robot at the operation speed of cooperation between others, when a high damping factor is set. In contrast, humans can perform an action quickly, whereas the stability of the position operation is poor if a low damping factor is set. In [5], the variable impedance control method instead of traditional impedance control is proposed, wherein the damping coefficient of impedance control can take two different values depending on the velocity of the robot.

In [6] and [7], the adaptive control method is put forward based on a neural network, which can reduce the complex model and does not need explicit parameters or functions. However, it is difficult to design the neural network framework and test the number of nodes to guarantee the network convergence. What is worse, the online computing speed of the neural network is too slow, hence making it hard to satisfy the real-time requirement of our experiments. In [8], the online variable impedance control is proposed, where the impedance model can be adjusted by force sensor information. This model is based on velocity control, which is more conventional than position control, and has better human–robot interaction in a low-level controller.

In particular, suitable parameters for the impedance control algorithm are difficult to be selected, even if the method has good performance with respect to the robot's operation. Notably, the main cause of the problem is that the impedance controller is typically related to many factors influencing its stability and

Manuscript received March 29, 2018; revised June 22, 2018 and November 1, 2018; accepted November 14, 2018. Date of publication December 7, 2018; date of current version May 31, 2019. This work was supported by the NSFC Shenzhen Robotics Research Center Project (U1613219), by the Shenzhen Overseas Innovation and Entrepreneurship Research Program (KQJSCX20170731164301774), and by the Shenzhen Technology Project (JSGG20160301160759264).
(Corresponding author: Xinyu Wu.)

Y. Hu is with the CAS Key Laboratory of Human–Machine Intelligence-Synergy Systems and Guangdong Provincial Key Lab of Robotics and Intelligent System, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen 518055, China, and also with the Department of Informatics, Technical University of Munich, Munich 85748, Germany (e-mail: yingbai.hu@hotmail.com).

X. Wu and P. Geng are with the CAS Key Laboratory of Human–Machine Intelligence-Synergy Systems and Guangdong Provincial Key Lab of Robotics and Intelligent System, Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen 518055, China (e-mail: xy.wu@siat.ac.cn; gengpeng02@baidu.com).

Z. Li is with the Department of Automation, University of Science and Technology of China, Hefei 230022, China (e-mail: zjli@ieee.org).

Color versions of one or more of the figures in this paper are available online at http://ieeexplore.ieee.org.

Digital Object Identifier 10.1109/TIE.2018.2884240

convergence speed. In addition to the deep understanding of designing the parameters of the controller model, the model of environment and robotic system should be considered [9]. In [10], Mitrovic *et al.* proposed a stochastic optimal controller based on a definitive model to learn variable impedance controller models, whereas the definitive model's derivation is not easy. Under the inspiration of Mitrovic *et al.*'s work, we present the Covariance Matrix Adaptation-Evolution Strategies (CMA-ES) algorithm for learning variable impedance controller.

As a branch of evolution strategies [11], CMA-ES is a stochastic search method for problem optimization [12]. It can efficiently minimize the non linear objective function, and especially it shows good performance in abnormal conditions and non separable problems. In Kern *et al.*'s research, CMA-ES algorithm outperforms the multi modal Rastrigin function [13], and it is very effective in solving real-value coded variable problems [14]. In contrast to the stochastic search method of model-based algorithm, the gain parameters of a variable impedance controller can be learned by CMA-ES, which is a conventional approach to obtain the desired parameters of a controller. Also, CMA-ES is a black-box optimization method that can optimize the objective tasks, even if the model of the robot system is unknown. Compared with [15], CMA-ES learning method is also adopted to learn gain schedules for variable impedance controller, even if the model of the robot system is unknown, which can be adjusted in real time for tasks requirements and ensure the safety when robot interact with environment or human.

In this paper, a novel scheme that combines the redundant solution with dynamical movements primitives (DMPs) is proposed. DMPs is a useful approach to model the planning trajectories [16], which is represented as a set of equations, and the detail is shown in Section IV-A. DMPs has many outstanding properties, including good robustness in perturbation environment, as well as stability and convergence of the dynamic system. The parameters $\boldsymbol{\theta}$ in DMPs determine the shape of the trajectories, and they are very suitable as learning parameters. In view of the dimensional space problem, DMPs obtains a single-dimension output trajectory $[y_t, \dot{y}_t, \ddot{y}_t]$, which is very convenient for the iterative learning method of updating the DMPs parameters.

In this paper, the vision system is used to get the coordinates of the target object in Cartesian space. Then, primal-dual neural network (PDNN) is adopted to solve the redundant degrees of freedom (DoFs) problem of the 4-DoFs manipulator that is subject to physical constraints [17], and obtain the planning trajectories in joint space. In our control strategies, the reference trajectories are modeled by DMPs with the optimization solutions. Subsequently, CMA-ES is applied to learn the trajectories model with perturbations by learning the DMPs' parameters, which determines the shape of the trajectories. The contributions can be summarized as follows.

1) A novel scheme for tasks learning is presented that transforms the Cartesian space learning problem into the joint space learning problem, where PDNN algorithm is applied to solve the redundancy resolution for online planning with the vision feedback in Cartesian space, and DMPs is used to model the planning trajectories in joint space.
2) CMA-ES is employed to learn the trajectories that are generated by DMPs in a learning process. After updates, the manipulator can reach the target location more accurately even with unknown disturbance. CMA-ES has good convergence performance with external disturbance.
3) CMA-ES is also adopted to learn gain schedules for variable impedance controller, even if the model of the robot system is unknown, which can be adjusted in real time for tasks requirements and ensure the safety when robot interact with environment or human. The learning process of shape and controller parameters shares a cost function in a learning system based on CMA-ES, which reduces the complexity of learning problems.

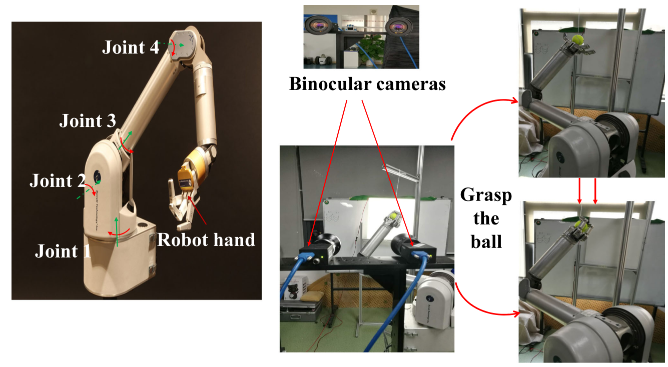


Fig. 1. Robot grasping system with stereo vision system.

II. Barrett WAM Robotic System

As shown in Fig. 1, the Barrett WAM robotic arm has four DoFs, including a BarrettHand with three fingers. The Barrett WAM has its own internal WAM PC, which executes a real-time Linux operating system. Compared with traditional robot arm, the robot arm has some good characteristics, such as high flexibility, zero backlash, and particularly, the near-zero friction between adjacent joints. The manipulator uses high-performance brushless motors to drive all joints. The WAM actuators are driven by Barrett's patented power modules, called Pucks, which are shown in Fig. 2.

A. Object Detection by the Vision System

The images of the ball are obtained by the stereo vision system using Blacklight, and the red segmented area in the images represents the ball. After getting the images by the stereo vision system, the image coordinates of the sphere center in both right and left images are extracted using the circle Hough Transform algorithm. Then, the three-dimensional (3-D) point position of the sphere center can be computed easily by triangulation [18]. The object ball in a system of Binocular cameras is presented in Fig. 3(b). Three coordinate systems of the Grasping System with vision are illustrated in Fig. 3(a), where o denotes as the

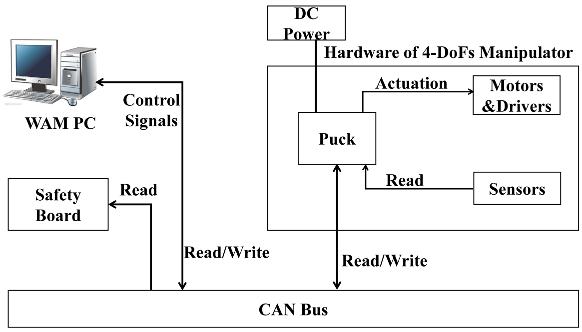


Fig. 2. System's overall hardware framework of WAM.

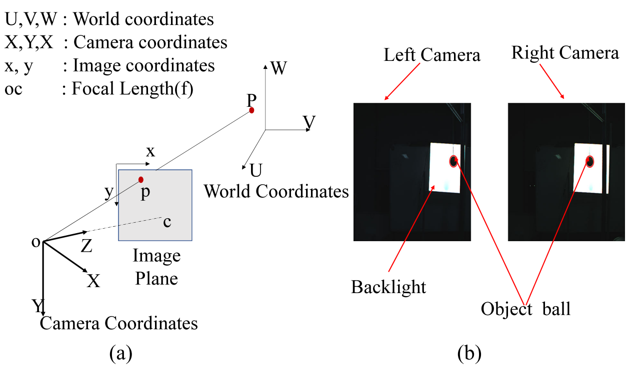


Fig. 3. Stereo vision system. (a) Vision Coordinates System. (b) The images of ball.

center of the camera and plane, p is the projection point from a 3-D point P onto the image plane. The pose matrix of ($\mathbf{R}$ and $\mathbf{T}$) between camera coordinate system and world (robot) coordinate system needs to be obtained.

According to Hartley and Zisserman's work [19], using the direct linear transform solution algorithm, the transform matrix ($\mathbf{R}$, $\mathbf{T}$) for an approximate estimate of the pose can be acquired, whereas the minimum value of the objective function cannot be obtained. A simple method of improving the result is changing the pose matrix of ($\mathbf{R}$ and $\mathbf{T}$) slightly in a random manner and then checking if the value of the error objective function decreases. Although a better solution can be gained with this method by adding disturbances to $\mathbf{R}$ and $\mathbf{T}$ until the objective function converges, the procedure is too slow to meet the experiment's requirements. In this paper, the Levenberg–Marquardt optimization [20] is employed to solve the problem.

III. Kinematics of Manipulator and Neurodynamics Optimization

A. Kinematics Redundant Resolution Formulation

The relationship between the end-effector position and the orientation variable $\boldsymbol{r} \in \mathbb{R}^m$, and joint variable $\boldsymbol{q} \in \mathbb{R}^n$ are expressed as follows:

$$\mathbb{J}(\boldsymbol{q})\dot{\boldsymbol{q}} = \dot{\boldsymbol{r}} \tag{1}$$

where $\dot{\boldsymbol{r}}$ is the velocity of the robot's end-effector position, and $\dot{\boldsymbol{q}}$ is the velocity of the joint position. The symbol $\mathbb{J}(\boldsymbol{q})$ is the Jacobian matrix.

We know there are infinitely many solutions of the redundant manipulator under the condition of $m < n$. The traditional pseudo inverse method has the high complexity of algorithms, and high analytical cost of solution. Considering the physical limits, the problem can be transformed as a convex optimization problem with conditional constraints [21]. The objective function is expressed as follows:

$$\text{minimize} \quad \frac{\dot{\boldsymbol{q}}^T\dot{\boldsymbol{q}}}{2} \tag{2}$$

$$\text{subject to} \quad \mathbb{J}\dot{\boldsymbol{q}} = \dot{\boldsymbol{r}} \tag{3}$$

$$\boldsymbol{q}^- \leqslant \boldsymbol{q}(t) \leqslant \boldsymbol{q}^+ \tag{4}$$

$$\dot{\boldsymbol{q}}^- \leqslant \dot{\boldsymbol{q}}(t) \leqslant \dot{\boldsymbol{q}}^+ \tag{5}$$

where $\mathbb{J}$ is the Jacobian matrix for the manipulator. By considering the physical limit, the lower and upper limits of the joint velocity and the joint position are denoted as $\boldsymbol{q}^-$ and $\boldsymbol{q}^+$ as well as $\dot{\boldsymbol{q}}^-$ and $\dot{\boldsymbol{q}}^+$, respectively. Furthermore, the boundaries of joint velocity meet the following inequality:

$$\varphi\left(\boldsymbol{q}^- - \boldsymbol{q}\right) \leq \dot{\boldsymbol{q}} \leq \varphi\left(\boldsymbol{q}^+ - \boldsymbol{q}\right) \tag{6}$$

where the parameter $\varphi > 0$ is scaled the feasible region of $\dot{q}$. In addition, the large value $\varphi > 0$ may cause fast joint deceleration when a joint approaches its limits, and φ should satisfy the condition: $\varphi \geq \max_{1\leq i\leq n}\left\{(\dot{\boldsymbol{q}}_i^+ - \dot{\boldsymbol{q}}_i^-)/(\boldsymbol{q}_i^+ - \boldsymbol{q}_i^-)\right\}$. For the detail proof of (6), see Appendix.

Therefore, we have the boundary of velocity: $\boldsymbol{\xi}_i^- = \max\left\{\dot{\boldsymbol{q}}_i^-, \varphi(\boldsymbol{q}_i^- - \boldsymbol{q}_i)\right\}$, $\boldsymbol{\xi}_i^+ = \min\left\{\dot{\boldsymbol{q}}_i^+, \varphi(\boldsymbol{q}_i^+ - \boldsymbol{q}_i)\right\}$; thus, the objective function can be reexpressed as

$$\text{minimize} \quad \frac{\dot{\boldsymbol{q}}^T\mathbb{W}\dot{\boldsymbol{q}}}{2} + \boldsymbol{b}^T\dot{\boldsymbol{q}} \tag{7}$$

$$\text{subject to} \quad \mathbb{J}\dot{\boldsymbol{q}} = \boldsymbol{l} \tag{8}$$

$$\boldsymbol{\xi}^- \leqslant \dot{\boldsymbol{q}} \leqslant \boldsymbol{\xi}^+ \tag{9}$$

where $\mathbb{W} = \boldsymbol{I}$, and $\boldsymbol{b} = 0$, $\boldsymbol{l} = \dot{\boldsymbol{r}}$. Overall, the objective function with the physical limit can be rewritten as the following quadratic programming (QP) problem

$$\text{minimize} \quad \frac{\mathcal{H}^T\mathbb{W}\mathcal{H}}{2} + \boldsymbol{b}^T\mathcal{H} \tag{10}$$

$$\text{subject to} \quad \mathbb{J}\mathcal{H} = \boldsymbol{l} \tag{11}$$

$$\boldsymbol{\xi}^- \leqslant \mathcal{H} \leqslant \boldsymbol{\xi}^+ \tag{12}$$

where the variable $\mathcal{H}$ denotes the joint velocity $\dot{\boldsymbol{q}}$, and $\boldsymbol{l}$ is the velocity of the end-effector $\dot{\boldsymbol{r}}$.

B. Neuro-Dynamics Optimization

To solve the QP problem depicted in (10)–(12) online, a PDNN is proposed that is based on a piecewise group of linear equations: linear variational inequalities (LVI) in [21] and [22]. In the study, the simplifying gradient LVI-PDNN is applied to solve the QP problem in (10)–(12). We define a dual decision

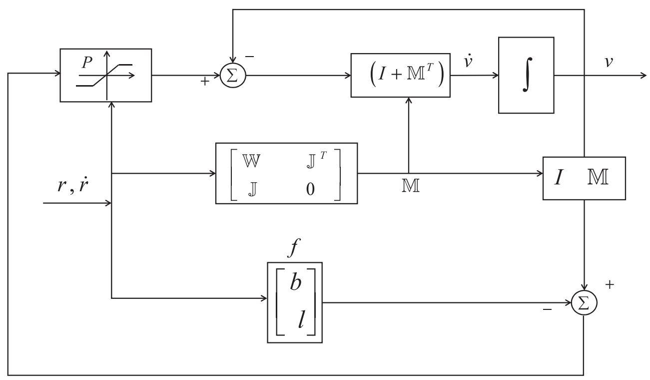


Fig. 4. Block diagram of LVI-PDNN.

vector $\boldsymbol{v} \in \Lambda = \{\boldsymbol{v}|\boldsymbol{v}^- \leq \boldsymbol{v} \leq \boldsymbol{v}^+\}$, which can be represented by the physical limit in (11)

$$\boldsymbol{v} = \begin{bmatrix} \mathcal{H} \\ \boldsymbol{y} \end{bmatrix}, \boldsymbol{v}^+ = \begin{bmatrix} \boldsymbol{\xi}^+ \\ \boldsymbol{y}^+ \end{bmatrix}, \boldsymbol{v}^- = \begin{bmatrix} \boldsymbol{\xi}^- \\ \boldsymbol{y}^- \end{bmatrix} \tag{13}$$

where $\boldsymbol{y}^+ \to +\infty$, $\boldsymbol{y}^- \to -\infty$, $(\boldsymbol{v}, \boldsymbol{v}^+, \boldsymbol{v}^-) \in \mathbb{R}^{n+m}$, and a convex set $\Lambda = \{\boldsymbol{v} \in \mathbb{R}^{n+m}|\boldsymbol{v}^- \leq \boldsymbol{v} \leq \boldsymbol{v}^+\} \subseteq \mathbb{R}^{n+m}$.

First, the QP problem (10)–(12) will be transformed into a set of LVIs.

Then, we need to choose a primal-dual equilibrium vector $\boldsymbol{v}^* \in \Lambda$ that meets the following condition:

$$(\boldsymbol{v} - \boldsymbol{v}^*)^T(\mathbb{M}\boldsymbol{v}^* + \boldsymbol{f}) \geqslant 0 \quad \forall \boldsymbol{v} \in \Lambda. \tag{14}$$

The variable matrix $\mathbb{M}$ and the vector $\boldsymbol{f}$ are denoted

$$\begin{bmatrix} \mathbb{W} & -\mathbb{J}^T \\ \mathbb{J} & \boldsymbol{0} \end{bmatrix} \in \mathbb{R}^{(n+m)\times(n+m)}, \boldsymbol{f} = \begin{bmatrix} \boldsymbol{b} \\ -\boldsymbol{l} \end{bmatrix} \in \mathbb{R}^{(n+m)}. \tag{15}$$

Second, we know the LVI (14) are equivalent to the system of piecewise linear equations [21], such as

$$P_\Lambda\left(\boldsymbol{v} - (\mathbb{M}\boldsymbol{v} + \boldsymbol{f})\right) - \boldsymbol{v} = 0 \tag{16}$$

where $P_\Lambda(\cdot)$ denotes the projection operator onto Λ, and each element is defined as

$$P_\Lambda(v_i) = \begin{cases} v_i^- & \text{if } v_i < v_i^- \\ v_i & \text{if } v_i^- \leqslant v_i \leqslant v_i^+ \\ v_i^+ & \text{if } v_i > v_i^+ \end{cases}.$$

It is obvious that we take the dynamic system to solve the LVI in (16), and (16) is formulated as

$$F(\boldsymbol{v}) = P_\Lambda\left(\boldsymbol{v} - (\mathbb{M}\boldsymbol{v} + \boldsymbol{f})\right) - \boldsymbol{v}. \tag{17}$$

To solve the QP problem in (10)–(12) if the dual variable v satisfies (17), we propose the following dynamic equation:

$$\dot{\boldsymbol{v}} = \gamma(\boldsymbol{I} + \mathbb{M}^T)F(\boldsymbol{v}) \tag{18}$$

where γ is a positive parameter that is designed to scale the convergence rate of the PDNN. The block diagram of LVI–PDNN is the real-time solution for the QP problem shown in Fig. 4. According to [21], the dual variable $\boldsymbol{v}(t)$ converges to an equilibrium point $\boldsymbol{v}^*$ from any initial state $\boldsymbol{v}(0)$.

IV. CMA-ES With Dynamical Movement Primitives

DMPs are a set of dynamic equations that are used to model the desired trajectories generated by a PDNN. In contrast, the CMA-ES algorithm is used to learn the reference trajectories with perturbations based on DMPs.

A. Dynamical Movement Primitives

DMPs is a predominant method for motor learning, especially in conjunction with other machine learning methods, such as reinforcement learning in [24]. DMPs are a set of dynamic system equations, whereas the core function (19) includes the linear spring damper part and the nonlinear part, which is defined as

$$\tau\dot{z}_t = \alpha_z(\beta_z(g - y_t) - z_t) + \boldsymbol{h}_t^T\boldsymbol{\theta} \tag{19}$$

$$\tau\dot{y}_t = z_t \tag{20}$$

$$\tau\dot{x}_t = \alpha_x x_t \tag{21}$$

$$\boldsymbol{h}_t = \frac{\sum_{i=1}^N \Psi_i(x_t)x_t}{\sum_{i=1}^N \Psi_i(x_t)}(g - y_0) \tag{22}$$

$$\Psi_i(x_t) = \exp\left(-\frac{1}{2\rho_i}(x_t - c_i)^2\right). \tag{23}$$

The transformation system (19)–(20) generate the reference trajectories $[y_t, \dot{y}_t, \ddot{y}_t]$, including the position, velocity, and acceleration, where g is the goal parameter of the attractor point, the vector $\boldsymbol{\theta}$ determines the shape of the attractor, (21) is the Canonical system in which x_t asymptotically decays to 0, and α_z, α_x, β_z, as well as τ are time constants. We can define the Gaussian basis functions $\boldsymbol{h}_t \in \mathbb{R}^{N\times1}(t \in [1, \ldots, N])$ with the bandwidth ρ_i and the center c_i of the Gaussian kernels. The shape parameter vector $\boldsymbol{\theta}$ determines the shape of robotic movement, e.g., the joint trajectory $[\ddot{y}_t, \dot{y}_t, y_t]$ over time.

Considering the danger of discontinuous jump $\ddot{y}$, we convert the g changing into a first-order differential equation

$$\tau\dot{g}_t = \alpha_g(G - g_t) \tag{24}$$

where g_t denotes parameter g, and the variable G denotes the discontinuous change of goal; so, g_t becomes the continuous parameter that can avoid the jump of acceleration. Although the modification converts the discontinuous parameter g into a continuous parameter, the scaling and stability characters of the system are the same as before [23].

To ensure the forcing term h_t generates a continuous value, we can adopt the same method of converting the first-order Canonical system into a second-order canonical system

$$\tau\dot{v}_t = \alpha_v(-\beta_v x_t - v_t) \tag{25}$$

$$\tau\dot{x}_t = v_t. \tag{26}$$

The forcing term is reformulated as

$$\boldsymbol{h}_t = \frac{\sum_{i=1}^N \Psi_i(x_t)v}{\sum_{i=1}^N \Psi_i(x_t)}(g_t - y_0). \tag{27}$$

The above DMPs system is the forward process, so we obtain the following equation according to (19) and (20):

$$f_{\text{target}} = (\boldsymbol{h}_t^T\boldsymbol{\theta})_{\text{target}} = \tau^2\ddot{y}_d - \alpha_z\left(\beta_z\left(g-y_d\right)-\tau\dot{y}_d\right). \quad (28)$$

The shape parameter of $\boldsymbol{\theta}$ can be obtained by a locally weighted regressed (LWR).

B. CMA-ES Algorithm

The evolutionary strategy (ES) can optimize a set of parameters by finding the optimization solution of search problems, such as the minimization of the nonlinear objective function in the search space. Generally, the approach of variables mutation or recombination is taken to decide the search steps, and the best search points are chosen as the new steps during the search process. In addition, the mutation is implemented by adding a Gaussian normally distributed random variable, which is the key part of the ES algorithm.

The samples meet the normal distribution in a (μ_α, m)-CMA-ES algorithm, while the new sample is determined by following distribution based on the old sample:

$$\boldsymbol{x}_k^{(\text{new})} \sim \mathcal{N}\left(\langle\boldsymbol{x}\rangle_\alpha, \sigma^2\boldsymbol{\Sigma}\right), k=1,\ldots m \quad (29)$$

where $\mathcal{N}(\langle\boldsymbol{x}\rangle_\alpha, \boldsymbol{\Sigma})$ is the Gaussian normally distributed random variable, $\langle\boldsymbol{x}\rangle_\alpha$ denotes the mean, and $\boldsymbol{\Sigma}$ is the covariance matrix. The recombination sample $\langle\boldsymbol{x}\rangle_\alpha = \sum_{i=1}^{\mu}\omega_i\boldsymbol{x}_{i:m}$ denotes the weighted mean of the selected individuals, and the variable ω_i satisfies the condition: $\sum_{i=1}^{\mu}\omega_i = 1(\omega_i > 0, i = 1\ldots\mu)$. In the CMA-ES algorithm, the index $i(i=1\ldots\mu)$ is the ith best individual of all samples.

Two major mutation variables are adopted in the learning steps, including the covariance matrix $\boldsymbol{\Sigma}$ as well as the global step size σ, whereas $\boldsymbol{\Sigma}$ is updated by the evolution path $\boldsymbol{p}_\Sigma^{\text{new}}$ as follows:

$$\boldsymbol{p}_\Sigma^{\text{new}} = (1-c_\Sigma)\boldsymbol{p}_\Sigma + h_\sigma\sqrt{c_\Sigma\left(2-c_\Sigma\right)\mu_p}\left(\delta\boldsymbol{x}/\sigma\right) \quad (30)$$

$$\boldsymbol{\Sigma}^{\text{new}} = \zeta\boldsymbol{\Sigma} + c_1\left(\boldsymbol{p}_\Sigma^{\text{new}}\boldsymbol{p}_\Sigma^{\text{new}T} + \delta(h_\sigma^{\text{new}})\Sigma\right) + c_\mu\mathcal{W} \quad (31)$$

$$\zeta = (1-c_1-c_\mu), \delta(h_\sigma^{\text{new}}) = (1-h(\delta^{\text{new}}))c_c(2-c_c)$$

$$\mathcal{W} = c_\mu\sum_{i=1}^{\mu}\frac{\omega_i}{\sigma^2}\left(\mathbf{x}_{i\ldots m}^{\text{new}} - \langle\mathbf{x}\rangle_\alpha\right)\cdot\left(\mathbf{x}_{i\ldots m}^{\text{new}} - \langle\mathbf{x}\rangle_\alpha\right)^T.$$

The vector $\delta\boldsymbol{x}$ denotes the difference between the new sample and previous $\langle\boldsymbol{x}\rangle_\alpha$: $\delta\boldsymbol{x} = \langle\boldsymbol{x}\rangle_\alpha^{\text{new}} - \langle\boldsymbol{x}\rangle_\alpha$, and the coefficient μ_p means the correct selection mass of variance based on Gaussian normally distribution: $\mu_p = 1/\sum_{i=1}^{\mu}\omega_i^2$ (especially when $\omega_i = 1/\mu$, and $\mu_p = \mu$). In (31), c_1 and c_μ are positive parameters that are used to scale the convergence rate of $\boldsymbol{\Sigma}$ in CMA-ES. The variable $\boldsymbol{p}_\sigma{}^{\text{new}}$ is the conjugate evolution path that determines the global-learning step size of the parameter $\boldsymbol{\sigma}^{\text{new}}$.

We define $\boldsymbol{\Sigma} = \boldsymbol{B}\cdot\boldsymbol{D}^2\cdot\boldsymbol{B}^T$ and then easily obtain the orthogonal matrix $\boldsymbol{B}$ along with the diagonal matrix $\boldsymbol{D}$ by analyzing its principal component. Therefore, the updated equation of parameter $\boldsymbol{p}_\sigma$ is expressed as

$$\boldsymbol{p}_\sigma^{\text{new}} = (1-c_\sigma)\boldsymbol{p}_\sigma + \sqrt{c_\sigma(2-c_\sigma)\mu_p}\cdot\boldsymbol{B}\boldsymbol{D}^{-1}\boldsymbol{B}^T\left(\delta\boldsymbol{x}/\sigma\right) \quad (32)$$

$$\boldsymbol{h}_\sigma^{\text{new}} = \begin{cases} 1, & \text{if } \frac{\|\boldsymbol{p}_\sigma^{\text{new}}\|}{\sqrt{1-(1-c_s)^2}} < k_1 E\left(\|\mathcal{N}(0,\boldsymbol{I})\|\right) \\ 0, & \text{otherwise} \end{cases} \quad (33)$$

where $k_1 = (1.5 + \frac{1}{n-0.5})$. The global step size σ^{new} obeys

$$\sigma^{\text{new}} = \sigma\cdot\exp\left(\frac{c_\sigma}{d_\sigma}\left(\frac{\|\boldsymbol{p}_\sigma^{\text{new}}\|}{E\left(\|\mathcal{N}(0,\boldsymbol{I})\|\right)} - 1\right)\right) \quad (34)$$

$$E\left(\|\mathcal{N}(0,\boldsymbol{I})\|\right) = \sqrt{2}\Gamma\left(\frac{n+1}{2}\right)/\Gamma\left(\frac{n}{2}\right) \approx \sqrt{n}\left(1-\frac{1}{4n}+\frac{1}{21n^2}\right)$$

where $E\left(\|\mathcal{N}(0,\boldsymbol{I})\|\right)$ is the desired value of $\boldsymbol{p}_\sigma$ under random selection. The initial values are set as Algorithm 1, and the details of coefficients' values in (30)–(34) are listed as

$$m = 4 + [3\ln(n)],$$

$$\mu = \left[\frac{m}{2}\right], \omega_{i=1\ldots\mu} = \frac{\ln(\mu+1)-\ln(i)}{\sum_{j=1}^{\mu}\ln(\mu+1)-\ln(i)}$$

$$c_\sigma = \frac{\mu_p+2}{n+\mu_p+3}, d_\sigma = 1+2\max\left(0,\sqrt{\frac{\mu_p-1}{n+1}}-1\right)+c_\sigma$$

$$c_c = \frac{4}{n+4}$$

$$c_1 = \frac{2}{(n+1.3)^2+\mu_p}, c_\mu = \min\left(1-c_1, 2\frac{\mu_p-2+1/\mu_p}{(n+2)^2+\mu_p}\right)$$

where d_σ denotes the damping coefficient for CMA-ES, and c_σ and $c_\sum$ represent the constant values of time. The above parameters can be adjusted by the user's choice.

C. Learning the Shape and Goal of DMPs Through CMA-ES

Typically, the shape parameters $\boldsymbol{\theta}$ are learned by imitation learning [25] and the policy parameters $\boldsymbol{\theta}$ are tuned such that the cost function can be minimized

$$S(\boldsymbol{\tau}_i) = \phi_{t_N} + \int_{t_i}^{t_N}\left(r_t + \frac{1}{2}\boldsymbol{\theta}^T\boldsymbol{R}\boldsymbol{\theta}\right)dt \quad (35)$$

where S denotes the cost function over all robotic trajectories that balance the performance of CMA-ES from the initial time to the end time(t_i to t_N). There are three parts in (35), including terminal cost ϕ_{t_N}, immediate cost r_t as well as immediate control cost $\frac{1}{2}\boldsymbol{\theta}_t^T\boldsymbol{R}\boldsymbol{\theta}_t$, and its expanded form is presented in Section V.

In this paper, the CMA-ES is also exploited to learn shape and goal parameters. The exploration noise ε is added to the

Algorithm 1: The CMA-ES Update Rule.

Given:
Initialize a cost function r_t, terminal cost ϕ_{t_N}, the variance $\boldsymbol{\Sigma}$ of the mean-zero noise $\boldsymbol{\Sigma}$, $\boldsymbol{\theta}$, initial state y_0, $\boldsymbol{p}_\sigma = 0$, $\boldsymbol{p}_\Sigma = 0$, covariance matrix $\boldsymbol{\Sigma} = \boldsymbol{I}$.
While cost not converged **do**
1. **For** $k = 1, \ldots, K$,
   - Sample exploration vector from Gaussian $\boldsymbol{\varepsilon}_k^\theta \sim \mathcal{N}(0, \sigma^2\boldsymbol{\Sigma})$, $\varepsilon_k^g \sim \mathcal{N}(0, \sigma^2\boldsymbol{\Sigma})$
   - Determine the cost of the sample $S_k = S(\boldsymbol{\theta} + \boldsymbol{\varepsilon}_k^\theta, G + \varepsilon_k^g)$
2. **Compute weights**(ε_k and $P_k \to (\boldsymbol{\varepsilon}_k^\theta, P_k^\theta)$, (ε_k^g, P_k^g))
   - $\varepsilon_{k=1\ldots K} \leftarrow$ **sort** $\varepsilon_{k=1\ldots K}$ w.r.t $S_{k=1\ldots K}$
   - **For** $k = 1, \ldots, K$,
     $h_\sigma = \begin{cases} \ln(\max(k/2, K_e) + 0.5) - \ln(k), & \text{if } k \le K_e \\ 0, & \text{if } k \le K_e \end{cases}$
3. **Update mean** $\delta\boldsymbol{\theta} = \sum_{k=1}^{K} P_k^\theta \boldsymbol{\varepsilon}_k^\theta$, $\delta G = \sum_{k=1}^{K} P_k^g \varepsilon_k^g$
4. **Update covariance matrix:** $\mathcal{W} = \sum_{k=1}^{K} P_k \boldsymbol{\varepsilon}_k \boldsymbol{\varepsilon}_k^T$
5. **Update parameters:** $\boldsymbol{\theta}^{\text{new}} = \boldsymbol{\theta} + \delta\boldsymbol{\theta}$, $G^{\text{new}} = G + dG$
6. **Update Covariance Matrix:**
   $\boldsymbol{p}_\sigma^{\text{new}} = (1 - c_\sigma)\boldsymbol{p}_\sigma + \sqrt{c_\sigma(2 - c_\sigma)\mu_p \boldsymbol{\Sigma}^{-1}}\mathcal{M}(\delta\boldsymbol{\theta}/\sigma)$
   $\boldsymbol{\sigma}^{\text{new}} = \sigma \cdot \exp\left(\frac{c_\sigma}{d_\sigma}\left(\frac{\|\boldsymbol{p}_\sigma\|}{E\|\mathcal{N}(0,\boldsymbol{I})\|} - 1\right)\right)$
   $\boldsymbol{p}_\Sigma^{\text{new}} = (1 - c_\Sigma)\boldsymbol{p}_\Sigma + h_\sigma\sqrt{c_\Sigma\,(2 - c_\Sigma)\,\mu_p}\,(\delta\boldsymbol{\theta}/\sigma)$
   $\boldsymbol{\Sigma}^{\text{new}} = (1 - c_1 - c_\mu)\boldsymbol{\Sigma} + c_1\left(\boldsymbol{p}_\Sigma \boldsymbol{p}_\Sigma^T + \delta(h_\sigma)\boldsymbol{\Sigma}\right) + c_\mu \mathcal{W}$
   $(\mathcal{M} = \boldsymbol{B} \cdot \boldsymbol{D}^{-1} \cdot \boldsymbol{B}^T)$

shape and goal parameters, which are represented by $\boldsymbol{\theta} + \boldsymbol{\varepsilon}_k^\theta$ and $G + \varepsilon_k^g$; then, the noise can be obtained by sampling from the zero-mean Gauss distribution as

$$\tau\ddot{y}_t = \alpha_z(\beta_z(g_t - y_t) - \dot{y}_t) + \boldsymbol{h}_t^T(\boldsymbol{\theta} + \boldsymbol{\varepsilon}_k^\theta) \tag{36}$$

$$\tau\dot{g}_t = \alpha_g(G + \varepsilon_k^g - g_t) \tag{37}$$

where the goal exploration noise ε^g comes from a Gaussian with variance $\sum^g$. The noise of $\varepsilon_{k=1,\ldots,K}^g$ is sampled at the initial epoch, and the number of samples K is set by the user. Our expectation is that the goal state g_t will converge immediately to $G + \varepsilon_k^g$ when the parameter α_g is set to a large value.

Every noisy DMP parameter can produce slightly different movements to lead to different costs, and $\{\boldsymbol{\tau}_t\}_k = \{\ddot{y}_{t\,i}, \dot{y}_{t\,i}, y_{t_i}\}_k$ is the set of K exploration trials.

The $\boldsymbol{\theta}$ of DMPs represents the $\langle \boldsymbol{x} \rangle_\alpha$ in (29). The parameter update is the most important part of the policy improvement loop in Algorithm 1. The updating law has robustness to the cost functions of discontinuity and noise, where the probabilistic weighted average is independent of the computational gradient. Both $\boldsymbol{\theta}$ and g_t are updated through the same costs, and the probability weights are similar. Negative interference does not exist between learning $\boldsymbol{\theta}$ and g_t.

V. Variable Impedance Control With CMA-ES

A. Variable Impedance Control With DMPs

As a classic robotic control method, negative feedback control with high proportional-derivative (PD) gains has such good features as easy implementation, robust modeling uncertainty, and low computational cost. However, control with high gains is not ideal for the tasks involving environmental interaction, such as force control tasks or locomotion.

Instead, impedance control is aimed at achieving specific robot impedance in either the joint space or the end-effector. Nevertheless, the problem of specifying target impedance has not been fully resolved. For simple tasks, appropriate impedance characteristics can be derived if the environment and the characteristics of a task are known *a priori*. Another advantage of variable impedance behavior in robots is the active safety due to soft "giving in" for both robots and the environment. The motor torque is calculated through a PD control law with the feed-forward control term $\boldsymbol{T}_x$

$$\boldsymbol{T} = -\boldsymbol{K}_p(\boldsymbol{q} - \boldsymbol{q}_d) - \boldsymbol{K}_d(\dot{\boldsymbol{q}} - \dot{\boldsymbol{q}}_d) + \boldsymbol{T}_x \tag{38}$$

$$\boldsymbol{K}_d = 2\sqrt{\boldsymbol{K}_p} \tag{39}$$

where $\boldsymbol{K}_p$ and $\boldsymbol{K}_d$ are the position gain matrices and velocity gain matrices, respectively. $\dot{\boldsymbol{q}}$ and $\boldsymbol{q}$ are velocities and joint positions, respectively. $\boldsymbol{q}_d$ and $\dot{\boldsymbol{q}}_d$ are, respectively, the desired velocities and joint positions. For example, feed-forward control may be from a calculated torque control component or an inverse dynamic control unit. The choice of the gains $\boldsymbol{K}_p$ and $\boldsymbol{K}_d$ can parameterize the impedance of a joint. Variable impedance control allows $\boldsymbol{K}_p$ to vary with the movement.

As mentioned in [26], this can be realized if the gain plane as is represented as an additional dimension in DMPs. The gain does not have a specific target value. Therefore, the proportional gain is not expressed as a convergent transformation system to g but is directly calculated through a function approximator

$$\boldsymbol{K}_{p,t} = \boldsymbol{h}_{t,k}^T\boldsymbol{\theta}_k \tag{40}$$

$$\boldsymbol{h}_{t,k} = \frac{\Psi_i(x_t)}{\sum_{i=1}^N \Psi_i(x_t)}. \tag{41}$$

This equation simulates the time process of position gain and is coupled to the DMPs (19). $\boldsymbol{K}_{p,t}$ is expressed by a basis function and is linearly related to the parameter $\boldsymbol{\theta}_k$.

B. Variable Impedance Control With CMA-ES

The robot receives feedback about joint torques, the end-effector position, and joint positions. After the trial, the immediate cost r_t in (35) denotes feedback to task achievement

$$r_t = \eta_1 d(\boldsymbol{X}_t) + \eta_2 \sum_{j=1}^{4}\left(\boldsymbol{K}_{p,t}^j - \boldsymbol{K}_{p,t}^{j,\min}\right) + \eta_3\left|\ddot{\boldsymbol{X}}_t\right| \tag{42}$$

where $d(\boldsymbol{X})$ is the distance from the end-effector to the line connecting the start and the end points of the movement. $\sum_{j=1}^4\left(\boldsymbol{K}_{p,t}^j - \boldsymbol{K}_{p,t}^{j,\min}\right)$ is total proportional gains on all joints.

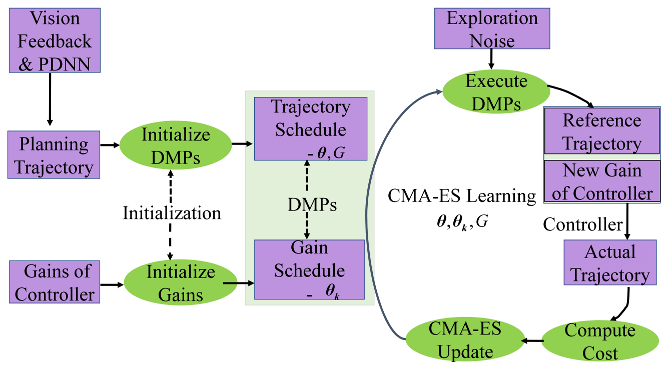


Fig. 5. Learning framework of the learning-based control system.

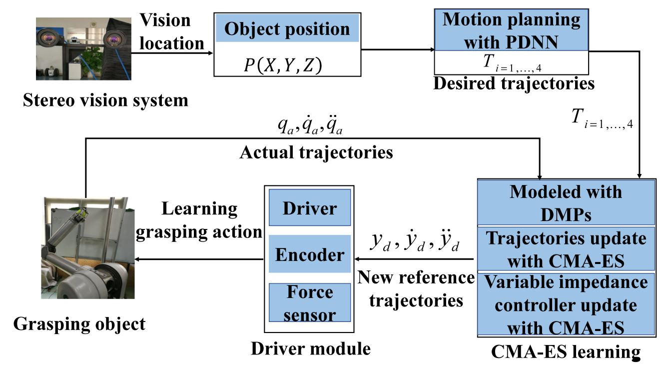


Fig. 6. Learning process of the grasping task.

$\left|\ddot{\boldsymbol{X}}_t\right|$ is the acceleration of the end-effector. When the Gaussian exploration is added to the gain, schedules similar to (36) and (40) can be expressed as

$$\boldsymbol{K}_{p,t} = \boldsymbol{h}_{t,k}^T \left(\boldsymbol{\theta}_k + \boldsymbol{\varepsilon}_k^{\boldsymbol{\theta}_k}\right). \tag{43}$$

Variable impedance control allows K_p to vary with the movement; this can be realized by expressing the gain schedules as extra dimensions in DMP, as discussed by [26]. The gain has no specific target value. Therefore, the proportional gain is directly calculated as the function approximator $\boldsymbol{K}_{p,t} = \boldsymbol{h}_{t,k}^T \left(\boldsymbol{\theta}_k + \boldsymbol{\varepsilon}_k^{\boldsymbol{\theta}_k}\right)$, as depicted in (43), instead of being expressed as a convergent transformation system to g. We learned the parameter of $\boldsymbol{\theta}_k$ with the same updated rule provided in Algorithm 1. The overall learning-based control strategy is illustrated in Fig. 5.

VI. Experiment

A. Experimental Setup

In this paper, the robot is operated to grasp the ball with the stereo visual system, and the grasping action is learned with CMA-ES algorithm. The task consists of two steps: first, online motion planning is carried out with visual system from the end-effector to the ball, where the visual system is applied to get the coordinates in Cartesian space, and PDNN is used to obtain the online planning trajectories of the redundant manipulator with the visual feedback information; second, the reference trajectories and gain schedules of variable impedance controller with perturbations are learned based on CMA-ES algorithm. The learning process of the robotic grasping task is shown in Fig. 6.

The parameters in (6) and (18) are set as: $\varphi = 4$, $\gamma = 6$. In practical applications, γ should not set too large to ensure that the convergence speed of network is not too slow. The goal parameters G_0 and other parameters of DMPs are presented in Table 1. The terminal cost ϕ_{t_N} in (35) for the task is set as

$$\phi_{t_N} = \eta_4 (y_t - g)^T (y_t - g) + \eta_5 \dot{y}_t^T \dot{y}_t \; (\tau \leq t \leq t_N).$$

The details of immediate cost r_t as well as the coefficient $(\eta_1, \ldots, \eta_5)$ in Cost function are also provided in Table 1. In addition, we set $t_N = 1.5\tau$ and keep the redundancy resolution as a constant value after τ seconds.

In our experiment, $\boldsymbol{\theta}_k$ is initialized by the locally weighted regression (LWR), and the values of the four joints $\boldsymbol{K}_p^{\min} = [100, 60, 16, 17]^T$ are given in Table I. In our control process, the variable impedance controller is employed to track the reference trajectories, which are generated by DMPs in each update. The real trajectories of the robot are used to compute the value of the reward (Cost value) in (35), which is used to update $\boldsymbol{\theta}_k$ of the variable impedance controller and update $\boldsymbol{\theta}$ of DMPs based on the CMA-ES algorithm in Algorithm 1. Then, the new $\boldsymbol{\theta}$ and $\boldsymbol{\theta}_k$ can generate the new variable impedance controller parameters $\boldsymbol{K}_p, \boldsymbol{K}_d$ and new reference trajectories $[y_t, \dot{y}_t, \ddot{y}_t]$, respectively. The details of the learning framework for learning-based control system are shown in Fig. 5. In our experiment, the number of samples is set as $\lambda = 6$, and the exploration noise was set as $\boldsymbol{\Sigma}^\theta = 1$, $\boldsymbol{\Sigma}^g = 1 \cdot l^u$ and $\boldsymbol{\Sigma}^{\boldsymbol{\theta}_k} = 10^{-1} \cdot \boldsymbol{K}_p^{\min}$, where $l = 0.9$ and u denotes the number of updates.

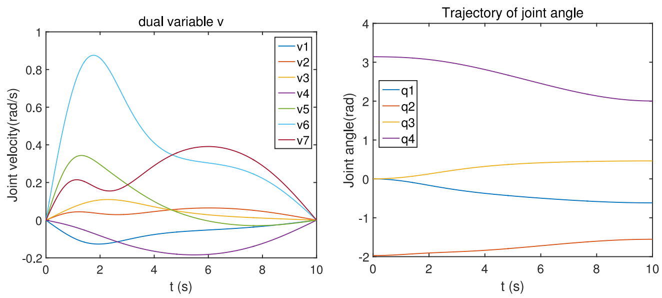
B. Result and Analysis

From the results of planning trajectories, which are illustrated in Fig. 7(a)–(b), it can observed that the planning trajectories are continuous and smooth, also showing the good convergence performance of the joint angles and velocity trajectories. Obviously, the results verify the effectiveness of the PDNN method on redundancy DoFs problems when physical constraint is considered.

The learning processes of all joints are shown in Figs. 8(a)–11(a), and the results indicate that the exploration trajectories gradually converge to the desired trajectories with the increase of updates. By comparison, all the samples' trajectories substantially coincide with the desired trajectories at the last updates, which are displayed in Figs. 8(b)–11(b), thus proving the learning performance of the CMA-ES algorithm while noises are added to the control system and reference trajectories. The gain schedules $\boldsymbol{K}_p$ of the variable impedance controller learning process and $\boldsymbol{K}_p$ updating at the last updates are illustrated in Figs. 12(a)–15(a), and Figs. 12(b)–15(b), respectively. It can be seen that $\boldsymbol{K}_p$ changing as the task requirements: increase quickly with high position error, and decrease with low

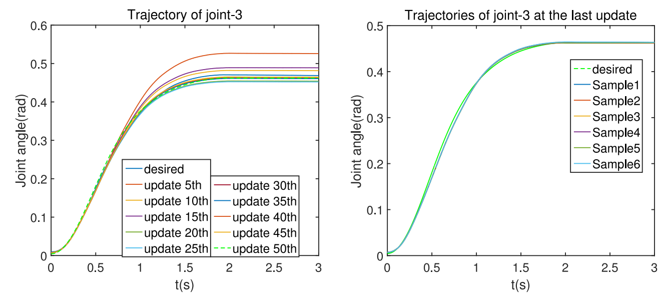
TABLE I
EXPERIMENT PARAMETERS AND RESULTS

Name	Parameters and results
Initial position of all joints	$q_0 = [-0.007; -1.994; 0.00456; 3.127]^T rad$
Position of the ball in Cartesian space	$T = [-0.2, 0.4, 0.43]^T m$
Optimization scheme in Eq.(6) and Eq.(18)	$\varphi = 4, \gamma = 6$
Parameters of DMPs systems	$\alpha_z = 25, \beta_z = 25/4, \alpha_g = 25/2, \alpha_x = 1, \alpha_v = \beta_v = 0.1, \tau = 2$
Goal parameters of DMPs	$G_0 = [-0.6164, -1.5527, 0.4625, 2.0035]^T rad$
Minimum of $\boldsymbol{K}_p$	$\boldsymbol{K}_p^{min} = [100, 60, 16, 17]^T$
Parameters of Cost function	$\eta_1 = 10^8, \eta_2 = 1, \eta_3 = 10^{-1}, \eta_4 = 10^3, \eta_5 = 10^3, \boldsymbol{R} = 10$



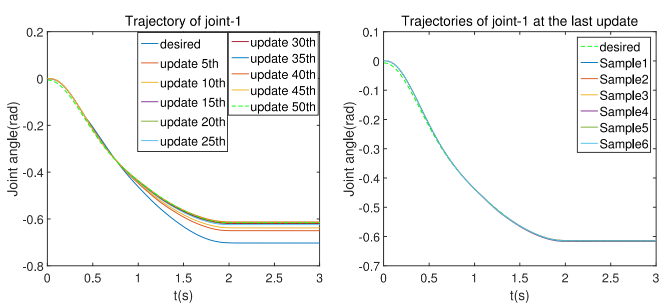
(a) (b)

Fig. 7. Redundancy-resolution with PDNN. (a) Redundancy-resolution. (b) Trajectory of joint angle.



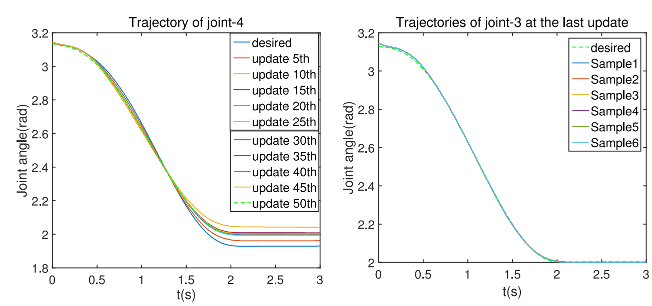
(a) (b)

Fig. 10. exp1: Joint-3 trajectories learning results. (a) One of samples' learning process. The actual trajectory gradually converge to the desired trajectory. (b) Trajectories at the last update of all samples.



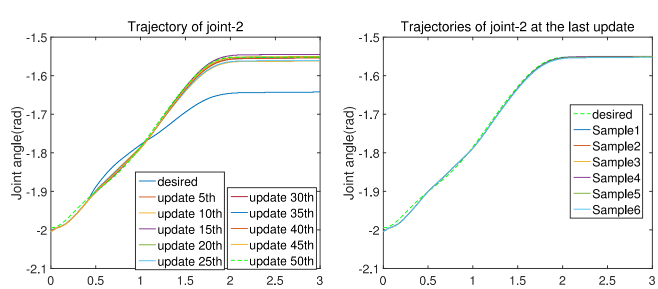
(a) (b)

Fig. 8. exp1: Joint-1 trajectories learning results. (a) One of samples' learning process. The actual trajectory gradually converge to the desired trajectory. (b) The trajectories at the last update of all samples.



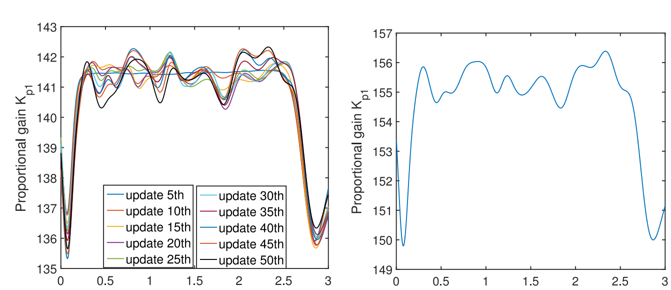
(a) (b)

Fig. 11. exp1: Joint-4 trajectories learning results. (a) One of samples' learning process. The actual trajectory gradually converge to the desired trajectory. (b) Trajectories at the last update of all samples.



(a) (b)

Fig. 9. exp1: Joint-2 trajectories learning results. (a) One of samples' learning process. The actual trajectory gradually converge to the desired trajectory. (b) Trajectories at the last update of all samples.



(a) (b)

Fig. 12. exp1: Proportional Gain K_{p1} learning results of controller. (a) Learning process of K_{p1}. (b) K_{p1} at the last update.

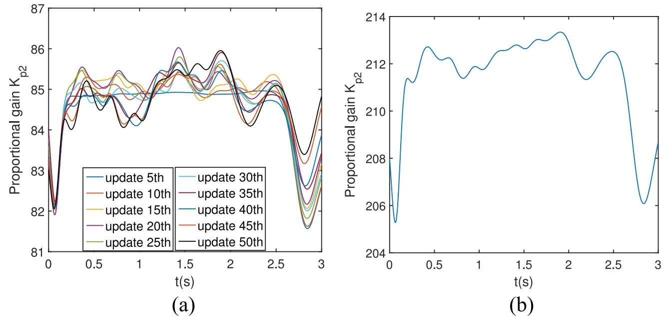


Fig. 13. exp1: Proportional Gain K_{p2} learning results of controller. (a) Learning process of K_{p2}. (b) K_{p2} at the last update.

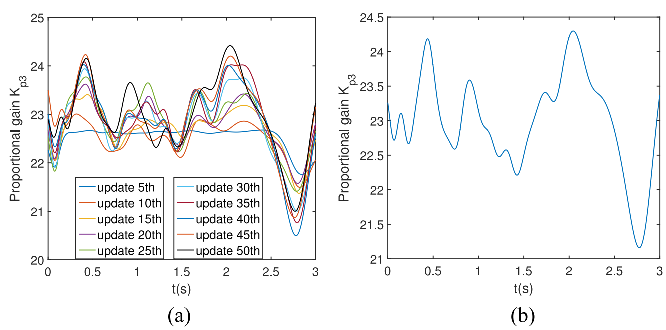


Fig. 14. exp1: Proportional Gain K_{p3} learning results of controller. (a) Learning process of K_{p3}. (b) K_{p3} at the last update.

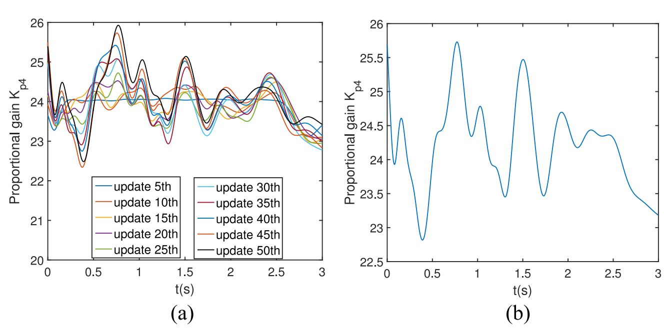


Fig. 15. exp1: Proportional Gain K_{p4} learning results of controller. (a) Learning process of K_{p4}. (b) K_{p4} at the last update.

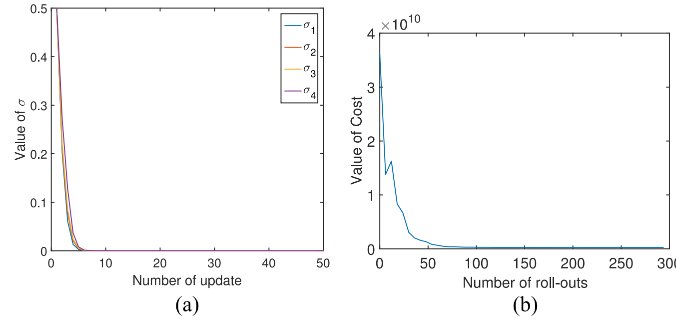


Fig. 16. exp1: Learning results of σ and cost value. (a) Parameter-σ learning of manipulator. (b) Cost of manipulator.

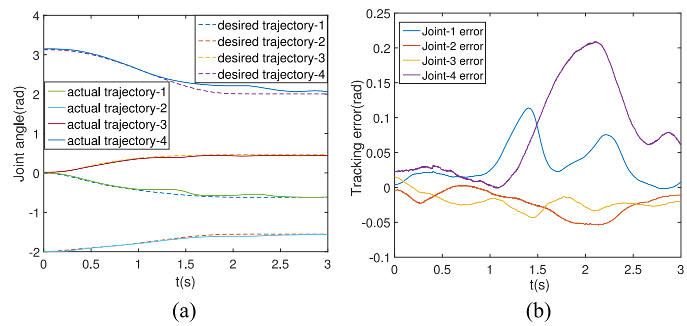


Fig. 17. Tracking the planning trajectories without CMA-ES. (a) Tracking trajectories with traditional impedance control. (b) Tracking errors.

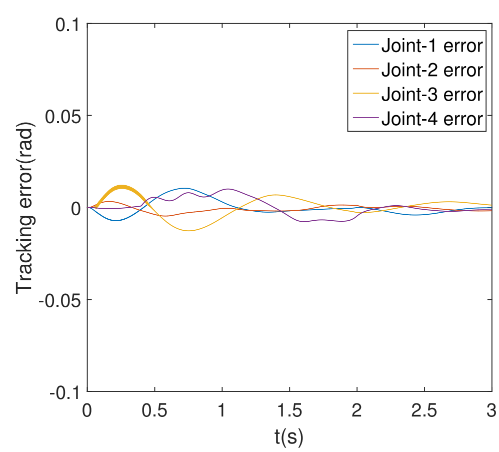


Fig. 18. Tracking error of manipulator.

position. After 50 updates, all the samples of $\boldsymbol{K}_p$ gradually converge to the last one.

The parameters of the σ learning process are illustrated in Fig. 16(a), which denotes that the global step size converges at a small value, and they demonstrate the go od search performance of CMA-ES algorithm. Besides, the cost function S gradually reduces and converges at a certain value that demonstrates the performance of the CAM-ES algorithm, also illustrated in Fig. 16(b). The tracking errors of the manipulator converge close to zero as shown in Fig. 17, suggesting that the variable impedance control based on CMA-ES can make the robot system track the desired trajectories.

The results are shown in Fig. 18(a)–(b), which represent the tracking result by traditional impedance control method. From the results in Figs. 17–18(b), it can be seen that the tracking errors are much higher than those of the variable impedance controller based on CMA-ES learning method when disturbances impact robotic system. Therefore, the results validate the effectiveness of the proposed methods.

In order to verify that the proposed approach can suppress uncertain disturbance, the random resistance from human hand is applied to manipulator after $t = 0.9$ s in experiment-2, which is shown in Fig. 19(a). The trajectories of the manipulator

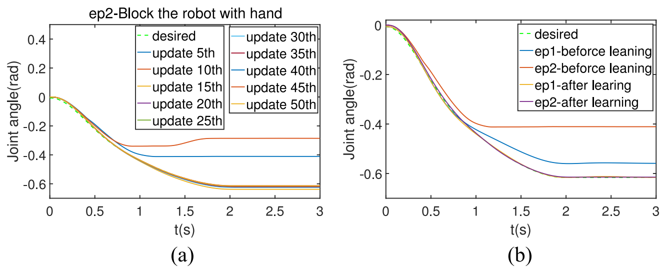


Fig. 19. exp2: Joint-1 trajectories. (a) Different disturbance experiment. The actual trajectory gradually converge to the desired trajectory. (b) Different disturbance experiment in learning process.

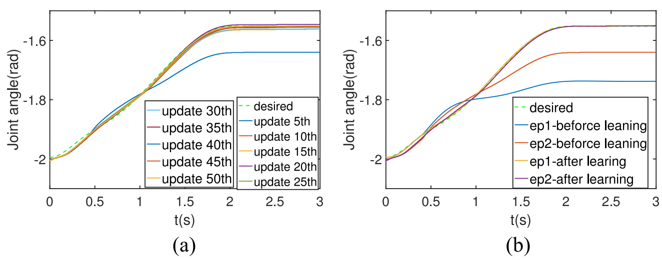


Fig. 20. exp2: Joint-2 trajectories. (a) Different disturbance experiment. The actual trajectory gradually converge to the desired trajectory. (b) Different disturbance experiment in learning process.

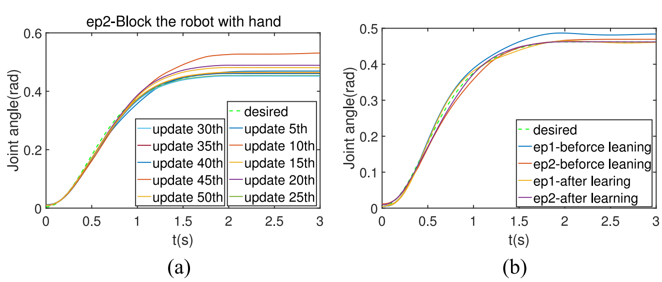


Fig. 21. exp2: Joint-3 trajectories. (a) Different disturbance experiment. The actual trajectory gradually converge to the desired trajectory. (b) Different disturbance experiment in learning process.

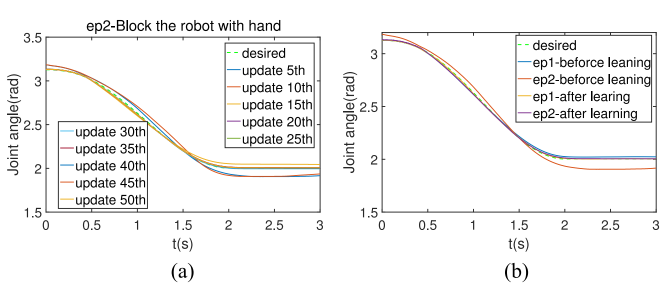


Fig. 22. exp2: Joint-4 trajectories. (a) Different disturbance experiment. The actual trajectory gradually converge to the desired trajectory. (b) Different disturbance experiment in learning process.

deviated from the desired which lead to a rapid increase in the cost function value. After several updates, the motion of manipulator gradually converge to the desired trajectory, and the cost function also converges to a small value by learning the shape, goal parameters, and gain schedules of controller. The results are shown in Fig. 19(b), and Figs. 20(a)–23(a). By comparing the different disturbances in the manipulator movement,

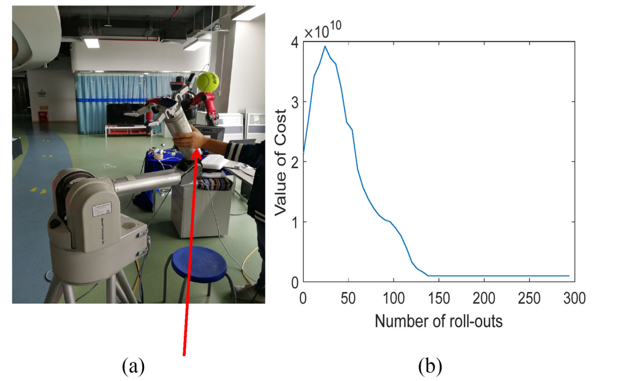


Fig. 23. exp2: Cost of manipulator with blocking. (a) Block the robot with hand. (b) The cost of manipulator.

the robot can suppress uncertain disturbance, and complete the learning tasks by the proposed learning methods. The results are shown in Figs. 19(b)–22(b).

VII. Conclusion

In this paper, the CMA-ES approach was presented for learning the motion skills of grasping under uncertain perturbations in conjunction with DMPs. In practice, the method was applied to learn the joint trajectories and parameters of a variable impedance controller by learning $\boldsymbol{\theta}$ and $\boldsymbol{\theta}_k$ based on DMPs. The proposed hybrid planning strategies were employed for robot operation, where DMPs incorporating PDNN were used to model the planning trajectories between the joint space and the task space, and CMA-ES incorporating DMPs was used to learn the reference trajectories and variable impedance controller with uncertain perturbations. Furthermore, the trajectories of gain schedules of impedance control indicate the expected behavior, and the parameters of $\boldsymbol{K}_p$ change as desired: it will increase with the task requirements, and stay the low-gain in other cases. Compared with the traditional approaches, the results of our experiment on the redundance robot have demonstrated that CMA-ES can learn the variable impedance controller without a model of the dynamic equations or the robot model system.

Appendix
Proof of Joint limits in (6)

We set the $\triangle t$ as the sampling time. Therefore, the joint physical limits at the next time step can be denoted as

$$\boldsymbol{q}_i^- \leq \boldsymbol{q}_i(t+\Delta t) \leq \boldsymbol{q}_i^+ \tag{44}$$

$$\dot{\boldsymbol{q}}_i^- \leq \dot{\boldsymbol{q}}_i(t+\Delta t) \leq \dot{\boldsymbol{q}}_i^+. \tag{45}$$

In (44), the joint velocity $\dot{\boldsymbol{q}}_i(t+\Delta t)$ could be approximated

$$\dot{\boldsymbol{q}}_i(t+\Delta t) \approx \frac{\boldsymbol{q}_i(t+\Delta t) - \boldsymbol{q}_i(t)}{\Delta t}.$$

The above equation can be reformulated as

$$\boldsymbol{q}_i(t+\Delta t) \approx \dot{\boldsymbol{q}}_i(t+\Delta t)\Delta t + \boldsymbol{q}_i(t).$$

The joint position limit in (44) can be expressed as

$$\boldsymbol{q}_i^- \leq \dot{\boldsymbol{q}}_i(t+\Delta t)\Delta t + \boldsymbol{q}_i(t) \leq \boldsymbol{q}_i^+.$$

Then, we can get the following:

$$\frac{\boldsymbol{q}_i^- - \boldsymbol{q}_i(t)}{\Delta t} \leq \dot{\boldsymbol{q}}_i(t+\Delta t) \leq \frac{\boldsymbol{q}_i^+ - \boldsymbol{q}_i(t)}{\Delta t}.$$

Therefore, (44)–(45) can be expressed as

$$\begin{cases} \frac{\boldsymbol{q}_i^- - \boldsymbol{q}_i(t)}{\Delta t} \leq \dot{\boldsymbol{q}}_i(t+\Delta t) \leq \frac{\boldsymbol{q}_i^+ - \boldsymbol{q}_i(t)}{\Delta t} \\ \dot{\boldsymbol{q}}_i^- \leq \dot{\boldsymbol{q}}_i(t+\Delta t) \leq \dot{\boldsymbol{q}}_i^+ \end{cases}.$$

Then, set $\varphi = 1/\triangle t$, and we will get the following:

$$\begin{cases} \varphi\left(\boldsymbol{q}_i^- - \boldsymbol{q}_i(t)\right) \leq \dot{\boldsymbol{q}}_i(t+\Delta t) \leq \varphi\left(\boldsymbol{q}_i^+ - \boldsymbol{q}_i(t)\right) \\ \dot{\boldsymbol{q}}_i^- \leq \dot{\boldsymbol{q}}_i(t+\Delta t) \leq \dot{\boldsymbol{q}}_i^+ \end{cases}.$$

So, the above physical limits can be converted into (6).

References

[1] D. Wang, P. Huang, and Z. Meng, "Coordinated stabilization of tumbling targets using tethered space manipulators," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 51, no. 3, pp. 2420–2432, Jul. 2015.

[2] Y. Zhang, P. Huang, Z. Meng, and Z. Liu, "Precise angles-only navigation for noncooperative proximity operation with application to tethered space robot," *IEEE Trans. Control Syst. Technol.*, 2018, doi: 10.1109/TCST.2018.2790400.

[3] P. Huang *et al.*, "Dexterous tethered space robot: Design, measurement, control, and experiment," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 53, no. 3, pp. 1452–1468, Jun. 2017.

[4] P. K. Jamwal, S. Hussain, M. H. Ghayesh, and S. V. Rogozina, "Impedance control of an intrinsically compliant parallel ankle rehabilitation robot," *IEEE Trans. Ind. Electron.*, vol. 63, no. 6, pp. 3638–3647, Jun. 2016.

[5] F. Ficuciello, L. Villani, and B. Siciliano, "Variable impedance control of redundant manipulators for intuitive human–robot physical interaction," *IEEE Trans. Robot.*, vol. 31, no. 4, pp. 850–863, Aug. 2015.

[6] Z. Li *et al.*, "Adaptive impedance control for an upper limb robotic exoskeleton using biological signals," *IEEE Trans. Ind. Electron.*, vol. 64, no. 2, pp. 1664–1674, Feb. 2017.

[7] Z. Lu, P. Huang, and Z. Liu, "Predictive approach for sensorless bimanual teleoperation under random time delays with adaptive fuzzy control," *IEEE Trans. Ind. Electron.*, vol. 65, no. 3, pp. 2439–2448, Mar. 2018.

[8] V. Duchaine and C. M. Gosselin, "General model of human-robot cooperation using a novel velocity based variable impedance control," in *Proc. 2nd Joint EuroHaptics Conf., Symp. Haptic Interfaces Virtual Environ. Teleoperator Syst.*, 2007, pp. 446–451.

[9] B. Siciliano, L. Sciavicco, L. Villani, and G. Oriolo, *Robotics-Modelling, Planning and Control*. London, U.K.: Springer, 2009.

[10] D. Mitrovic, S. Klanke, M. Howard, and S. Vijayakumar, "Exploiting sensorimotor stochasticity for learning control of variable impedance actuators," in *Proc. IEEE-RAS Int. Conf. Humanoid Robots*, 2010, pp. 536–541.

[11] H. P. Schwefel, *Evolution and Optimum Seeking*. New York, NY, USA: Wiley-Interscience, 1995.

[12] N. Hansen and A. Ostermeier, "Completely derandomized self-adaptation in evolution," *Strategies, Evolu. Comput.*, vol. 9, no. 2, pp. 159–195, 2001.

[13] S. Kern, S. D. Muller, N. Hansen, D. Buche, J. Ocenasek, and P. Koumoutsakos, "Learning probability distributions in continuous evolutionary algorithms-a comparative review," *Natural Comput.*, vol. 3, no. 1, pp. 77–112, 2004.

[14] N. Hansen and S. Kern, "Evaluating the CMA evolution strategy on multimodal test functions," in *Proc. Int.Conf. Parallel Problem Solving From Nature*, 2004, vol. 8, pp. 282–291.

[15] Z. Li, T. Zhao, F. Chen, Y. Hu, C. Y. Su, and T. Fukuda, "Reinforcement learning of manipulation and grasping using dynamical movement primitives for a humanoid-like mobile manipulator," *IEEE/ASME Trans. Mechatronics*, vol. 23, no. 1, pp. 121–131, Feb. 2018.

[16] F. Stulp, E. A. Theodorou, and S. Schaal, "Reinforcement learning with sequences of motion primitives for robust manipulation," *IEEE Trans. Robot.*, vol. 28, no. 6, pp. 1360–1370, Dec. 2012.

[17] Y. Hu *et al.*, "Development of sensory-motor fusion-based manipulation and grasping control for a robotic hand-eye system," *IEEE Trans. Syst., Man, Cybernet., Syst.*, vol. 47, no. 7, pp. 1169–1180, Jul. 2017.

[18] Z. Yu, X. Guo, H. Lin, A. Lumsdaine, and J. Yu, "Line assisted light field triangulation and stereo matching," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2013, pp. 2792–2799.

[19] R. Hartley and A. Zisserman, *Multiple View Geometry in Computer Vision*. Cambridge, U.K.: Cambridge Univ. Press, 2003.

[20] K. Christian, Y. Nobuo, and F. Masao, "Levenberg-Marquardt methods with strong local convergence properties for solving nonlinear equations with convex constraints," *J. Comput. Appl. Math.*, vol. 172, no. 2, pp. 375–97, 2004.

[21] Y. Zhang, S. S. Ge, and T. H. Lee, "A unified quadratic-programming-based dynamical system approach to joint torque optimization of physically constrained redundant manipulators," *IEEE Trans. Syst., Man, Cybern. B, Cybern*, vol. 34, no. 5, pp. 1083–4419, Oct. 2004.

[22] D. Chen and Y. Zhang, "A hybrid multi-objective scheme applied to redundant robot manipulators," *IEEE Trans. Autom. Sci. Eng.*, vol. 14, no. 3, pp. 1337–1350, Jul. 2017.

[23] A. Ijspeert, J. Nakanishi, P. Pastor, H. Hoffmann, and S. Schaal, "Dynamical movement primitives: Learning attractor models for motor behaviors," *Neural Comput.*, vol. 25, no. 2, pp. 328–373, 2013.

[24] E. Theodorou, J. Buchli, and S. Schaal, "A generalized path integral control approach to reinforcement learning," *J. Mach. Learn. Res.*, vol. 11, no. 11, pp. 3137–3181, 2010.

[25] S. Schaal, A. Ijspeert, and A. Billard, "Computational approaches to motor learning by imitation," *Philosophical Trans. Roy. Soc. B: Biol. Sci.*, vol. 358, pp. 537–547, 2003.

[26] J. Buchli *et al.*, "Learning variable impedance control," *Int. J. Robot. Res.*, vol. 30, no. 7, pp. 820–833, 2011.

Yingbai Hu received the B.S. degree in automation from the Hubei University of Technology, Hubei, China, in 2014, and the M.S degree in control engineering from the South China University of Technology, Guangzhou, China, in 2017. He is currently working toward the Ph.D. degree in computer science from the Faculty of Informatics, Technical University of Munich, Munich, Germany.

His research interests include robot control, motion planning, neural network optimization, and robot learning.

Xinyu Wu (M'14) received the B.S. and M.S. degrees in automation from the Department of Automation, University of Science and Technology of China, Hefei, China, in 2001 and 2004, respectively, and the Ph.D. degree in robotics from the Chinese University of Hong Kong, Hong Kong, in 2008.

He is currently a Professor with the Center for Intelligent and Biomimetic systems, Shenzhen Institutes of Advanced Technology, Shenzhen, China. His research interests include intelligent robots and computer vision.

Peng Geng received the M.S. degree in control engineering from the University of Chinese Academy of Sciences, Beijing, China, in 2018.

Currently, he is a Research and Development Engineer with the Department of Autonomous Driving Technology, Baidu Inc., Beijing, China. His research interests include autonomous driving, robot learning, and machine learning.

Zhijun Li (M'07–SM'09) received the Ph.D. degree in mechatronics from Shanghai Jiao Tong University, Shanghai, China, in 2002.

From 2003 to 2005, he was a Postdoctoral Fellow with the Department of Mechanical Engineering and Intelligent Systems, University of Electro-Communications, Tokyo, Japan. From 2005 to 2006, he was a Research Fellow with the Department of Electrical and Computer Engineering, National University of Singapore, and Nanyang Technological University, Singapore. Since 2012, he has been a Professor with the College of Automation Science and Engineering, South China University of Technology, Guangzhou, China. Since 2017, he has been a Professor with the Department of Automation, University of Science and Technology, Hefei, China. His current research interests include service robotics, teleoperation systems, nonlinear control, neural network optimization, etc.

Dr. Li has been a Co-Chair of Technical Committee on Biomechatronics and Bio-robotics Systems (B^2S) since 2016, IEEE Systems, Man and Cybernetics Society, Technical Committee on Neuro-Robotics Systems, and IEEE Robotics and Automation Society. He is serving as an Editor-at-Large with the *Journal of Intelligent & Robotic Systems*, and as an Associate Editor of several IEEE Transactions. He has been the General Chair and Program Chair of 2016 and 2017 IEEE Conference on Advanced Robotics and Mechatronics, respectively.